

SYLLABUS

Full Syllabus of all five selected Courses.

1. Introduction to Machine Learning
2. Data Structures
3. Operating Systems
4. Database Management Systems
5. Foundation of Data Science

Course Contents and Lecture Schedule

No	Topic	No. of Lectures
1	INTRODUCTION TO MACHINE LEARNING	
1.1	Mock Test on Module 1, Module 2 and Module 3	1 hour
1.2	Mock Test on Module 4 and Module 5	1 hour
1.3	Feedback and Remedial class	1 hour
2	DATA STRUCTURES	
2.1	Mock Test on Module 1, Module 2 and Module 3	1 hour
2.2	Mock Test on Module 4 and Module 5	1 hour
2.3	Feedback and Remedial class	1 hour
3	OPERATING SYSTEMS	
3.1	Mock Test on Module 1 and Module 2	1 hour
3.2	Mock Test on Module 3, Module 4 and Module 5	1 hour
3.3	Feedback and Remedial class	1 hour
4	DATABASE MANAGEMENT SYSTEMS	
4.1	Mock Test on Module 1, Module 2 and Module 3	1 hour
4.2	Mock Test on Module 4 and Module 5	1 hour

4.3	Feedback and Remedial class	1 hour
5	FOUNDATIONS OF DATA SCIENCE	
5.1	Mock Test on Module 1, Module 2 and Module 3	1 hour
5.2	Mock Test on Module 4 and Module 5	1 hour
5.3	Feedback and Remedial class	1 hour

Model Question Paper**QP CODE:****Reg No:** _____**Name:** _____**PAGES :7****APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY****SIXTH SEMESTER B.TECH DEGREE EXAMINATION, MONTH & YEAR****Course Code: ADT308****Course Name: Comprehensive Course Work****Max. Marks: 50****Duration: 1 Hour****Objective type questions with multiple choices. Mark one correct answer for each question.****Each Question Carries 1 Mark**

- Application of machine learning methods to large databases is called
 - Data Mining
 - Artificial Intelligence
 - Big Data Computing
 - Internet of Things
- If machine learning model output involves target variable, then that model is called as
 - Descriptive Model
 - Predictive Model
 - Reinforcement Learning
 - All of the above
- In what type of learning labelled training data is used
 - Unsupervised Learning
 - Supervised Learning
 - Reinforcement Learning
 - Active Learning
- In following type of feature selection method we start with empty feature set

- (A) Forward Feature Selection (B) Backward Feature Selection
(C) Both A and B (D) None of the above
5. Which of the following is the best machine learning method?
- (A) Scalable (B) Accuracy
(C) Fast (D) All of the above
6. Data used to build a data mining model.
- (A) Training data (B) Validation data
(C) Test data (D) Hidden data
7. You are given reviews of few netflix series marked as positive, negative and neutral. Classifying reviews of a new netflix series is an example of
- (A) Supervised learning (B) Unsupervised learning
(C) Semisupervised learning (D) Reinforcement learning
8. Following are the types of supervised learning
- (A) Classification (B) Regression
(C) subgroup discovery (D) all of the above
9. The output of training process in machine learning is
- (A) machine learning model (B) machine learning algorithm
(C) null (D) accuracy
10. PCA is
- (A) forward feature selection (B) backward feature selection
(C) feature extraction (D) all of the above
11. Consider the following sequence of operations on an empty stack.
push(22); push(43); pop(); push(55); push(12); s=pop();
Consider the following sequence of operations on an empty queue.
enqueue(32); enqueue(27); dequeue(); enqueue(38); enqueue(12); q=dequeue();
The value of s+q is _____
- (A) 44 (B) 54 (C) 39 (D) 70
12. A B-tree of order (degree)5 and of height 3 will have a minimum of ____ keys.
- A. 624
B. 249
C. 124
D. 250

13. Construct a binary search tree by inserting 8, 6, 12, 3, 10, 9 one after another. To make the resulting tree as AVL tree which of the following is required?
- (A) One right rotation only
 (B) One left rotation followed by two right rotations
 (C) One left rotation and one right rotation
 (D) The resulting tree itself is AVL
14. In a complete 4-ary tree, every internal node has exactly 4 children or no child. The number of leaves in such a tree with 6 internal nodes is:
- (A) 20 (B) 18 (C) 19 (D) 17
15. Select the postfix expression for the infix expression $a+b-c+d*(e/f)$.
- (A) $ab+c-d+e*f/$ (B) $ab+c-def/*+$
 (C) $abc-+def/*+$ (D) $ab+c-def/*+$
16. Consider a hash table of size seven, with starting index zero, and a hash function $(2x + 5) \bmod 7$. Assuming the hash table is initially empty, which of the following is the contents of the table when the sequence 1, 4, 9, 6 is inserted into the table using closed hashing? Note that '_' denotes an empty location in the table.
- (A) 9, _, 1, 6, _, _, 4 (B) 1, _, 6, 9, _, _, 4
 (C) 4, _, 9, 6, _, _, 1 (D) 1, _, 9, 6, _, _, 4
17. **Compute the time complexity of the following function:**
- ```
void function(int n)
{
 int count = 0;
 for (int i=n/2; i<=n; i++)
 for (int j=1; j<=n; j = j + 2)
 for (int k=1; k<=n; k = k * 2)
 count++;
}
```
- A.  $O(n^2 \log n)$   
 B.  $O(n \log^2 n)$   
 C.  $O(n^3)$   
 D.  $O(n \log n^2)$
18. How many distinct binary search trees can be created out of 6 distinct keys?
- (A) 7 (B) 36 (C) 140 (D) 132
19. Which tree traversal performed on a binary search tree, results in ascending order listing of the keys?
- A. Pre-order  
 B. In-order  
 C. Post-order  
 D. Level-order

20. You are given pointers to first and last nodes of a singly linked list, which of the following operations are dependent on the length of the linked list?
- (A) Delete the first element
  - (B) Insert a new element as a first element
  - (C) Add a new element at the end of the list
  - (D) Delete the last element of the list
21. Suppose a disk has 400 cylinders, numbered from 0 to 399. At some time the disk arm is at cylinder 58, and there is a queue of disk access requests for cylinder 66, 349, 201, 110, 38, 84, 226, 70, 86. If Shortest-Seek Time First (SSTF) is being used for scheduling the disk access, the request for cylinder 86 is serviced after servicing \_\_\_\_\_ number of requests.
- (A) 1
  - (B) 2
  - (C) 3
  - (D) 4
22. If frame size is 4KB then a paging system with page table entry of 2 bytes can address \_\_\_\_\_ bytes of physical memory.
- (A)  $2^{12}$
  - (B)  $2^{16}$
  - (C)  $2^{18}$
  - (D)  $2^{28}$
23. Calculate the internal fragmentation if page size is 4KB and process size is 103KB.
- (A) 3KB
  - (B) 4KB
  - (C) 1KB
  - (D) 2KB
24. Which of the following scheduling policy is likely to improve interactivensess?
- (A) FCFS
  - (B) Round Robin
  - (C) Shortest Process Next
  - (D) Priority Based Scgeduling
25. Consider the following program
- Semaphore X=1, Y=0
- |                                                                                                   |                                                                                                                 |
|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <pre>Void A ( ) {     While (1)     {         P(X);         Print'1';         V(Y);     } }</pre> | <pre>Void B ( ) {     While (1)     {         P(Y);         P(X);         Print'0';         V(X);     } }</pre> |
|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
- The possible output of the program:
- (A) Any number of 0's followed by any number of 1's.
  - (B) Any number of 1's followed by any number of 0's.
  - (C) 0 followed by deadlock
  - (D) 1 followed by deadlock
26. In a system using single processor, a new process arrives at the rate of 12 processes per minute and each such process requires 5 seconds of service time. What is the percentage of CPU utilization?

- (A) 41.66                      (B) 100.00                      (C) 240.00                      (D) 60.00
27. A system has two processes and three identical resources. Each process needs two resources to proceed. Then  
 (A) Deadlock is possible                      (B) Deadlock is not possible  
 (C) Starvation may be present                      (D) Thrashing
28. Which of the following is true with regard to Round Robin scheduling technique?  
 (A) Responds poorly to short process with small time quantum.  
 (B) Works like SJF for larger time quantum  
 (C) Does not use a prior knowledge of burst times of processes.  
 (D) Ensure that the ready queue is always of the same size.
29. Thrashing can be avoided if  
 (A) the pages, belonging to working set of programs, are in main memory  
 (B) the speed of CPU is increased  
 (C) the speed of I/O processor is increased  
 (D) none of the above
30. The circular wait condition can be prevented by  
 (A) using thread  
 (B) defining a linear ordering of resource types  
 (C) using pipes  
 (D) all of the above
31. Let E1, E2 and E3 be three entities in an E/R diagram with simple single-valued attributes. R1 and R2 are two relationships between E1 and E2, where R1 is one-to-many, R2 is many-to-many. R3 is another relationship between E2 and E3 which is many-to-many. R1, R2 and R3 do not have any attributes of their own. What is the minimum number of tables required to represent this situation in the relational model?  
 (A) 3                      (B) 4                      (C) 5                      (D) 6
32. Identify the minimal key for relational scheme R(U, V, W, X, Y, Z) with functional dependencies  $F = \{U \rightarrow V, V \rightarrow W, W \rightarrow X, VX \rightarrow Z\}$   
 (A) UV                      (B) UW                      (C) UX                      (D) UY
33. It is given that: “Every student need to register one course and each course registered by many students”, what is the cardinality of the relation say “Register” from the “Student” entity to the “Course” entity in the ER diagram to implement the given requirement.  
 (A) M:1 relationship                      (B) M:N relationship  
 (C) 1:1 relationship                      (D) option (B) or(C)
34. Consider the relation branch( branch\_name, assets, branch\_city)  
 SELECT DISTINCT T.branch\_name FROM branch T, branch S WHERE T.assets>L.assets AND S.branch\_city = "TVM".  
 Finds the names of  
 (A) All branches that have greater assets than all branches located in TVM.

- (B) All branches that have greater assets than some branch located in TVM.  
 (C) The branch that has the greatest asset in TVM.  
 (D) Any branch that has greater asset than any branch located in TVM.

35. Consider the following relation instance, where “A” is primary Key.

| A1 | A2 | A3 | A4   |
|----|----|----|------|
| 1  | 1  | 1  | Null |
| 5  | 2  | 5  | 1    |
| 9  | 5  | 13 | 5    |
| 13 | 13 | 9  | 15   |

Which one of the following can be a foreign key that refers to the same relation?

- (A) A2 (B) A3 (C) A4 (D) ALL

36. A relation R(ABC) is having the tuples(1,2,1),(1,2,2),(1,3,1) and (2,3,2). Which of the following functional dependencies holds well?

- (A)  $A \rightarrow BC$  (B)  $AC \rightarrow B$  (C)  $AB \rightarrow C$  (D)  $BC \rightarrow A$

37. Consider a relation R with attributes A, B, C, D and E and functional dependencies  $A \rightarrow BC$ ,  $BC \rightarrow E$ ,  $E \rightarrow DA$ . What is the highest normal form that the relation satisfies?

- (A) BCNF (B) 3 NF (C) 2 NF (D) 1 NF

38. For the given schedule S, find out the conflict equivalent schedule.

S : r1(x); r2(Z); r3(X); r1(Z); r2(Y); r3(Y); W1(X); W2(Z); W3(Y); W2(Y)

- (A)  $T1 \rightarrow T2 \rightarrow T3$  (B)  $T2 \rightarrow T1 \rightarrow T3$   
 (C)  $T3 \rightarrow T1 \rightarrow T2$  (D) Not conflict serializable

39. Specialization is \_\_\_\_\_ process.

- (A) top-down (B) bottom up  
 (C) Both (A) and (B) (D) none of these

40. If D1, D2, ..., Dn are domains in a relational model, then the relation is a table, which is a subset of

- (A)  $D1 + D2 + \dots + Dn$  (B)  $D1 \times D2 \times \dots \times Dn$   
 (C)  $D1 \cup D2 \cup \dots \cup Dn$  (D)  $D1 - D2 - \dots - Dn$

41. For each value of the \_\_\_\_, the distribution of the dependent variable must be normal.

- (A) Independent variable (B) Dependent variable  
 (C) Intermediate variable (D) None of the mentioned above

42. Data Analytics uses \_\_\_\_ to get insights from data.

- (A) Statistical figures (B) Numerical aspects  
 (C) Statistical methods (D) None of the mentioned above

43. Linear Regression is the supervised machine learning model in which the model finds the best fit \_\_\_\_ between the independent and dependent variable.

- (A) Linear line
- (B) Nonlinear line
- (C) Curved line
- (D) All of the mentioned above

44. Amongst which of the following is / are the types of Linear Regression,

- (A) Simple Linear Regression
- (B) Multiple Linear Regression
- (C) Both A and B
- (D) None of the mentioned above

45. Amongst which of the following is / are the true about regression analysis?

- (A) Describes associations within the data
- (B) Modeling relationships within the data
- (C) Answering yes/no questions about the data
- (D) All of the mentioned above

46. The process of quantifying data is referred to as \_\_\_\_.

- (A) Decoding
- (B) Structure
- (C) Enumeration
- (D) Coding

47. Data Analysis is a process of,

- (A) Inspecting data
- (B) Data Cleaning
- (C) Transforming of data
- (D) All of the mentioned above

48. Least Square Method uses \_\_\_\_.

- (A) Linear polynomial
- (B) Linear regression
- (C) Linear sequence
- (D) None of the mentioned above

49. What is a hypothesis?

- (A) A statement that the researcher wants to test through the data collected in a study



(B) A research question the results will answer

(C) A theory that underpins the study

(D) A statistical method for calculating the extent to which the results could have happened by chance

50. \_\_\_\_ are used when we want to visually examine the relationship between two quantitative variables.

(A) Bar graph

(B) Scatterplot

(C) Line graph

(D) Pie chart

| QNo | Ans. Key | QNo | Ans. Key | QNo | Ans. Key | QNo | Ans. Key | QNo | Ans. Key |
|-----|----------|-----|----------|-----|----------|-----|----------|-----|----------|
| 1   | (A)      | 11  | (C)      | 21  | (C)      | 31  | (C)      | 41  | (A)      |
| 2   | (B)      | 12  | (B)      | 22  | (D)      | 32  | (A)      | 42  | (C)      |
| 3   | (B)      | 13  | (A)      | 23  | (C)      | 33  | (D)      | 43  | (A)      |
| 4   | (A)      | 14  | (C)      | 24  | (B)      | 34  | (C)      | 44  | (C)      |
| 5   | (D)      | 15  | (D)      | 25  | (D)      | 35  | (B)      | 45  | (B)      |
| 6   | (A)      | 16  | (D)      | 26  | (B)      | 36  | (D)      | 46  | (C)      |
| 7   | (A)      | 17  | (A)      | 27  | (B)      | 37  | (D)      | 47  | (D)      |
| 8   | (D)      | 18  | (D)      | 28  | (C)      | 38  | (B)      | 48  | (B)      |
| 9   | (A)      | 19  | (B)      | 29  | (A)      | 39  | (C)      | 49  | (A)      |
| 10  | (C)      | 20  | (D)      | 30  | (B)      | 40  | (D)      | 50  | (A)      |